

Olyvia E. Witham

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EDUCATION

Bachelor of Science, **Electrical and Computer Engineering**

May 2027

Embedded Systems and Computer Architecture

Robotics Minor

The University of Texas at Austin

Overall GPA: 3.74/4.00

Relevant Coursework: **Embedded and Real-Time Systems Lab**, Robot Mechanism Design, **Automatic Control**, **Linear System and Signals**, Gateway to Robotics, **Power Electronics Lab**, Algorithms, Engineering Communications, **Software Design and Implementation**, **Circuit Theory**, **Digital Logic Design**, Computer Architecture

SKILLS

Technical Skills:

- **C, Python, C++, Java, Verilog**, Assembly, SQL
- **MATLAB, PLECS, LTSpice, KiCad, Git**, Microsoft 365, Linux, MQTT, **OnShape, Altium**
- Interfacing **TI MSPM0G3507, ESP32, Arduino microcontrollers, Raspberry Pi, and FPGAs, C2000**
- **PCB design, Oscilloscope & Multimeter knowledge, Soldering**, 3D Printing, Laser Cutting

Soft Skills:

- Collaboration, communication, leadership, organizational skills, critical thinking/problem solving, adaptable

PROJECTS

Real-Time Operating System Race Robot, Embedded Systems Lab

Jan 2026 – Apr 2026

- Developed **autonomous race robot software** on a **dual-processor real-time operating system** using **C and Assembly on MSPM0**, enabling inter-processor communication via **CAN bus**
- Implemented **PWM-based motor control** with closed-loop **PID algorithms** to regulate motor speed and steering for precise motion control
- Integrated **infrared and TF-Luna distance sensors** to collect real-time environmental measurements for obstacle detection and navigation
- Designed multi-task RTOS architecture to coordinate sensing, control, and communication with deterministic timing constraints
- Implemented **Wi-Fi communication** using an **ESP microcontroller** to transmit telemetry data and enable remote monitoring and debugging

Programmable Stopwatch/Timer, Digital Logic Design

Apr 2025

- Designed **modular digital system** in **Verilog** implementing a **programmable stopwatch** and timer using a finite state machine (**FSM**) and datapath architecture
- Implemented **clock division, counters, and register logic** to control timekeeping and user inputs from pushbuttons and switches
- Interfaced **7-segment displays** and LEDs on a **Xilinx Basys3 FPGA**, mapping I/O pins using constraint files and verifying functionality on hardware
- Simulated and debugged designs using **Xilinx Vivado**, validating **sequential logic behavior** and resolving timing and state transition errors
- Integrated **multiple Verilog modules** into a complete **FPGA system** and demonstrated **real-time operation** on physical hardware

ChessBot, UT RAS Demobots

Jan 2024 – Present

- Programmed path planning software for the electromagnet and chess pieces using **Python**
- **Designed PCB** including a magnetometer, leds, i2c, and capacitors, with a team and tested with **oscilloscope**

- Established a messaging protocol between chess moves, path planning, and stepper controls using **MQTT**
- Developed virtual testing system for the chessboard using python scripts (matplotlib library)
- Interfaced ChessBot hardware using **Linux** and **Raspberry Pi** for control and data processing

Boost Converter for EBike, Power Electronics Lab

Aug 2025 – Dec 2025

- **Designed and soldered a 4-layer KiCad PCB** for a **24 V to 48 V DC-DC boost converter** using a **half-bridge topology**, achieving **over 95% efficiency**
- Implemented **PWM control and ADC sampling** in C on a **TI C2000 LaunchPad** to drive MOSFETs and monitor current/voltage
- Optimized **inductor value and switching frequency (140 kHz)** through **MATLAB analysis**, balancing efficiency, size, and thermal performance
- Applied **power electronics layout best practices** (minimized loop inductance, controlled parasitics, isolated power and signal paths) and **validated performance with oscilloscope-based testing**

Longhorn Crossing, University of Texas- Intro to Embedded Systems

Apr 2024

- **Constructed PCB** with a DAC, LCD, TI MSPM0G3507, button switches, potentiometer, and LEDs
- **Programmed game engine** in C and C++ with graphics for LCD, interrupts for sounds through DAC which correlated to LEDs, and input interrupts and ADC for game play from button switches and potentiometer

WORK EXPERIENCE

Electrical Engineering Intern, REV Robotics,

Aug 2026 - Present

Hardware Engineer, Accesswave Technologies,

May 2026 – Aug 2026

- **Design and test custom PCBs** for an **audio-to-haptic sensory** substitution system for Deaf and hard-of-hearing users
- Develop embedded **C++ firmware on ESP32** microcontrollers to interface sensors, actuators, and communication peripherals
- Integrate and debug hardware and firmware using **oscilloscopes, multimeters, and serial debugging to validate system performance**
- Collaborate with a multidisciplinary team to prototype, test, and refine assistive technology for real-world applications in fast paced startup environment

General Leader, IEEE Robotics and Automation Society,

Dec 2024– Present

- Assisted beginners in robotics with color sensing and line sensing
- Collaborated with leaders on issues of budget, meeting schedule, and projects
- Volunteered for tabling events to encourage new members